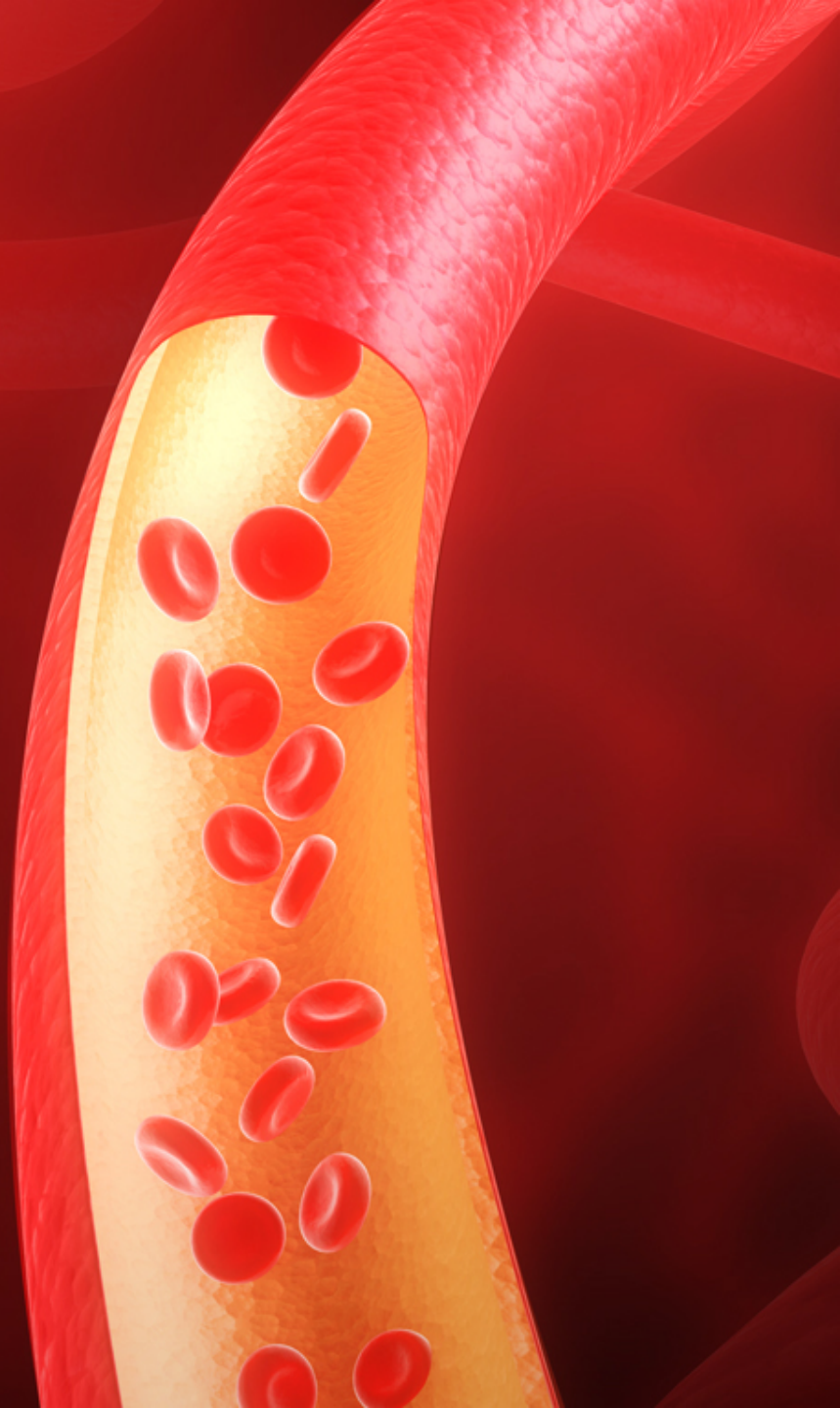
Assessment of Haemostatic Function





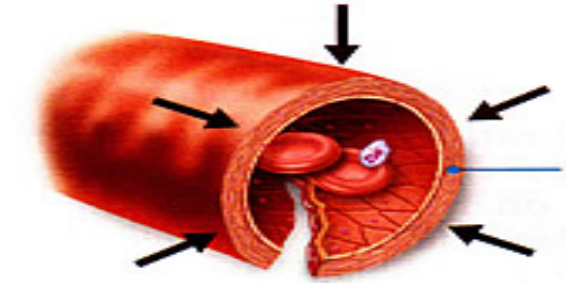
Haemostasis

- **Haemostasis:** Spontaneous Stoppage of bleeding from an injured blood vessel.

Stages of Haemostasis

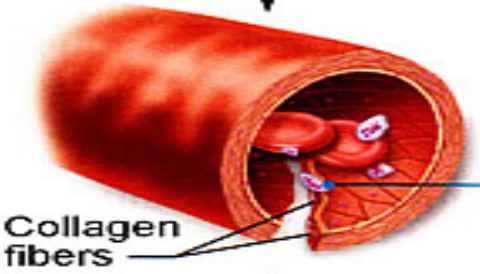
- ***Vascular Constriction***
- ***Formation of Platelet Plug***
- ***Formation of blood clot***
- ***Healing***





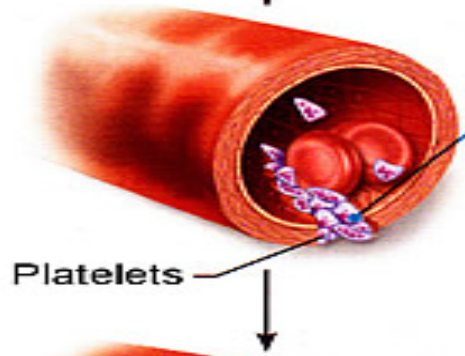
Step 1. Vascular spasm

- Contraction of the smooth muscle produces the vasoconstriction

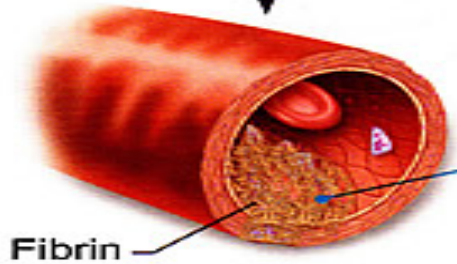


Step . Formation of the platelet plug

- Lesion of the endothelium exposes the collagen fibers; platelet adhesion is trigered



- Platelets release mediators which further favorize platelet aggregation; the platelet plug is formed



Step 3. The coagulation

- Fibrin filaments polymarize; red and white blood cells get trapped and the blood clot is formed



:Vasoconstriction- 1

- Mechanical
  (Local myogenic)
- Chemical
  (TXA2, Serotonin)
- Nervous
  (neurogenic reflex)

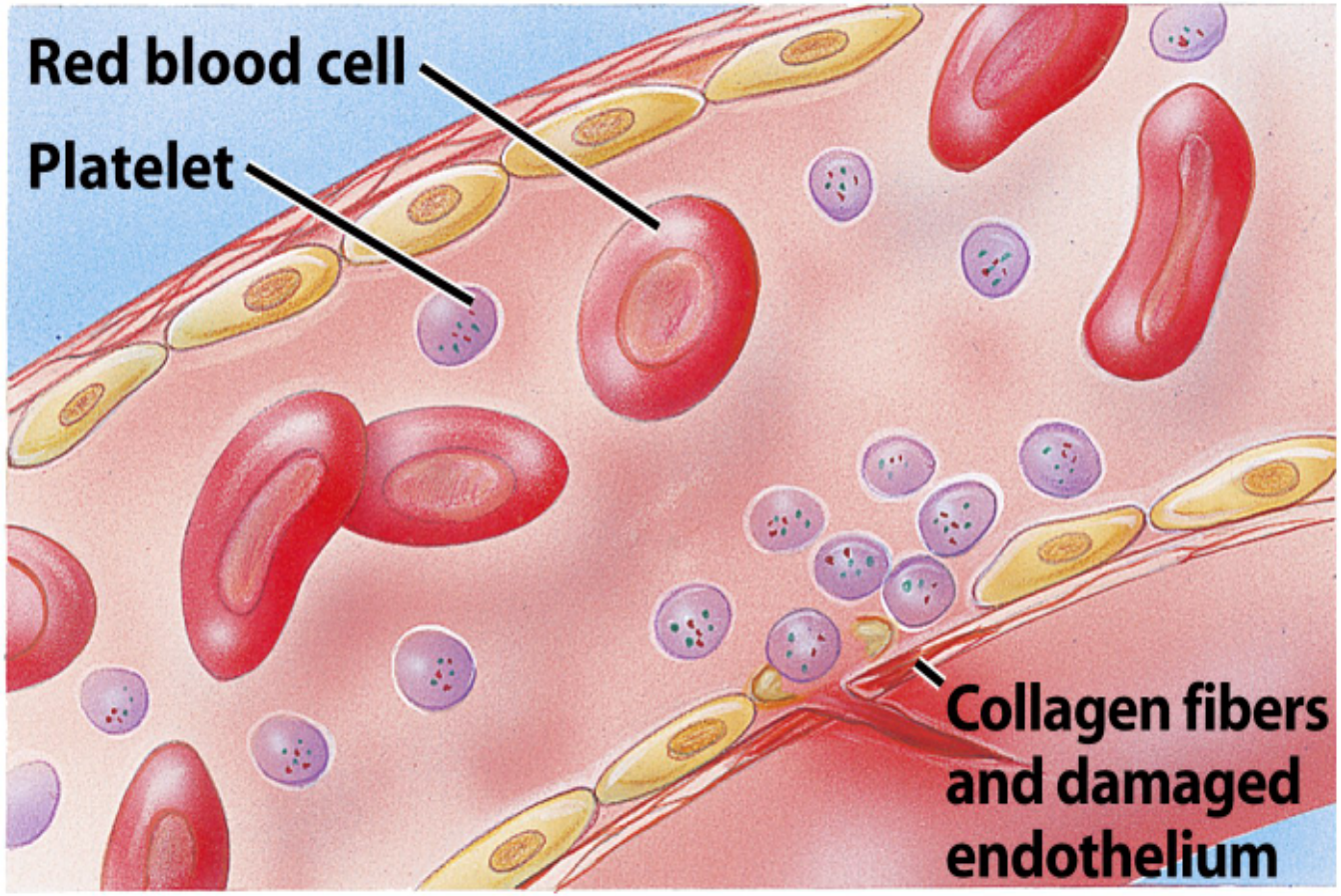


2- Platelet plug formation

(Temporary plug): Loose

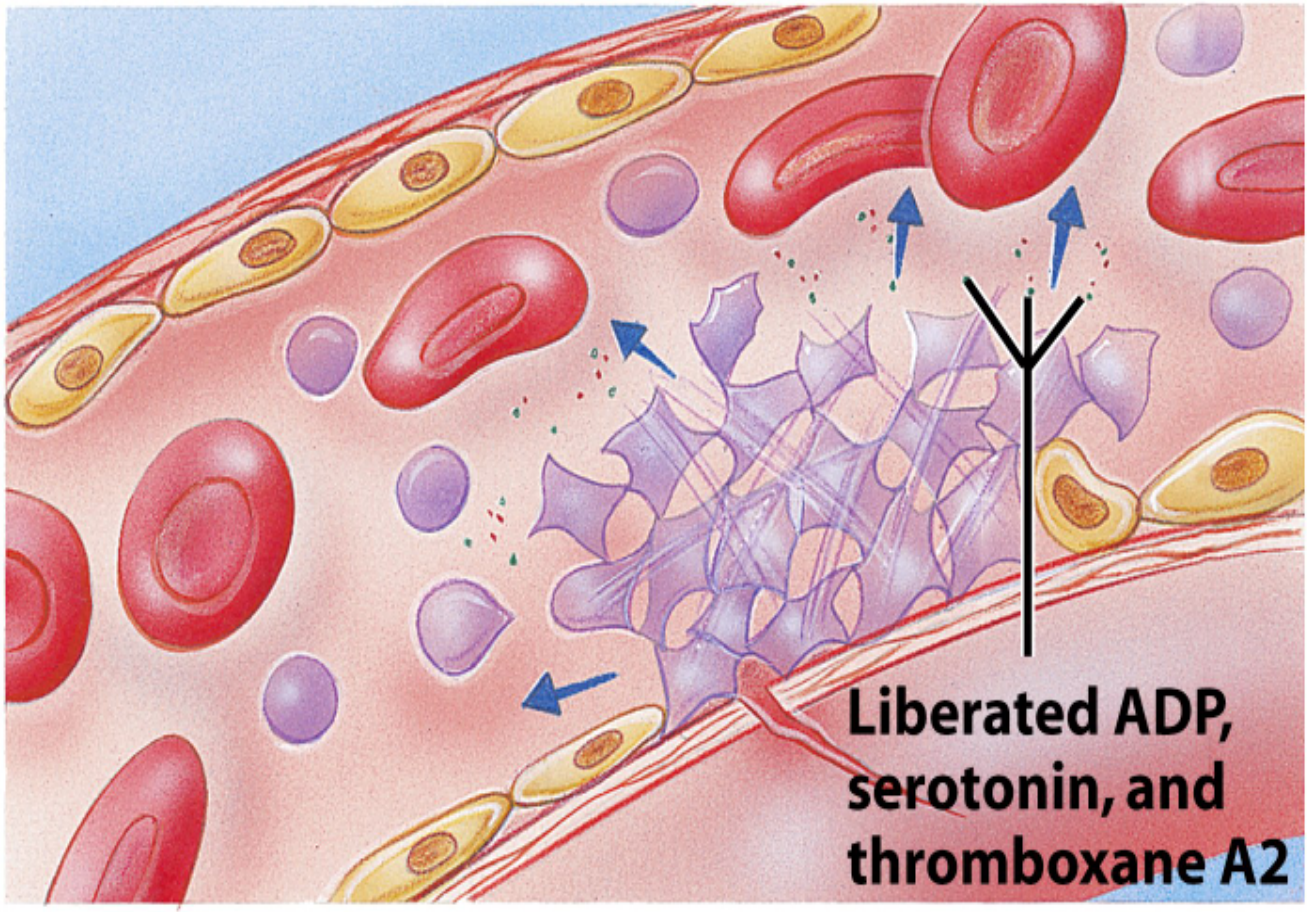
- *Adhesion (VWF)*
- *Secretion (ADP, TXA_2, PF_3)*
- *Aggregation(ADP)*



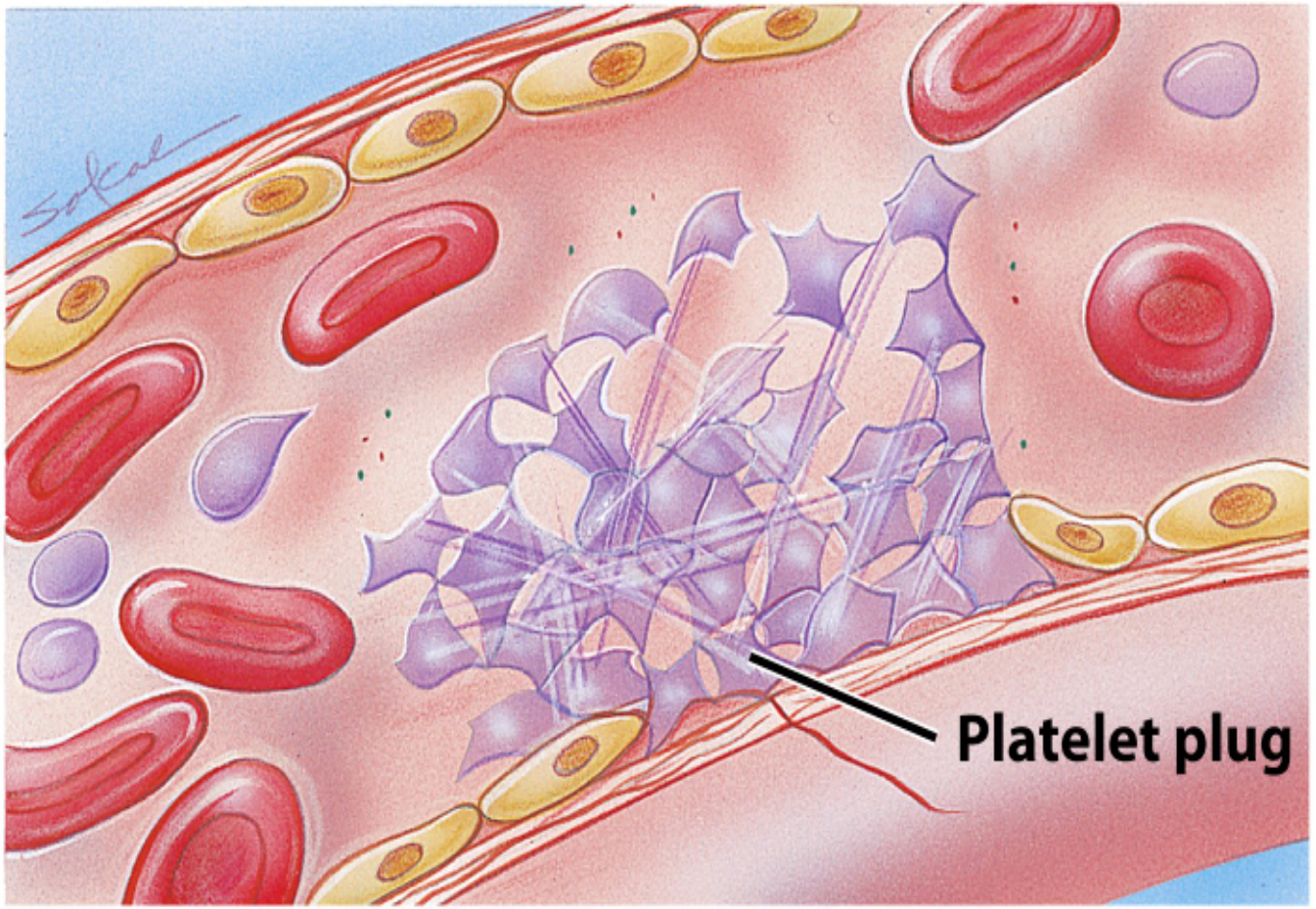


1 Platelet adhesion





2 Platelet release reaction



3 Platelet aggregation



Fibrin clot formation- 3

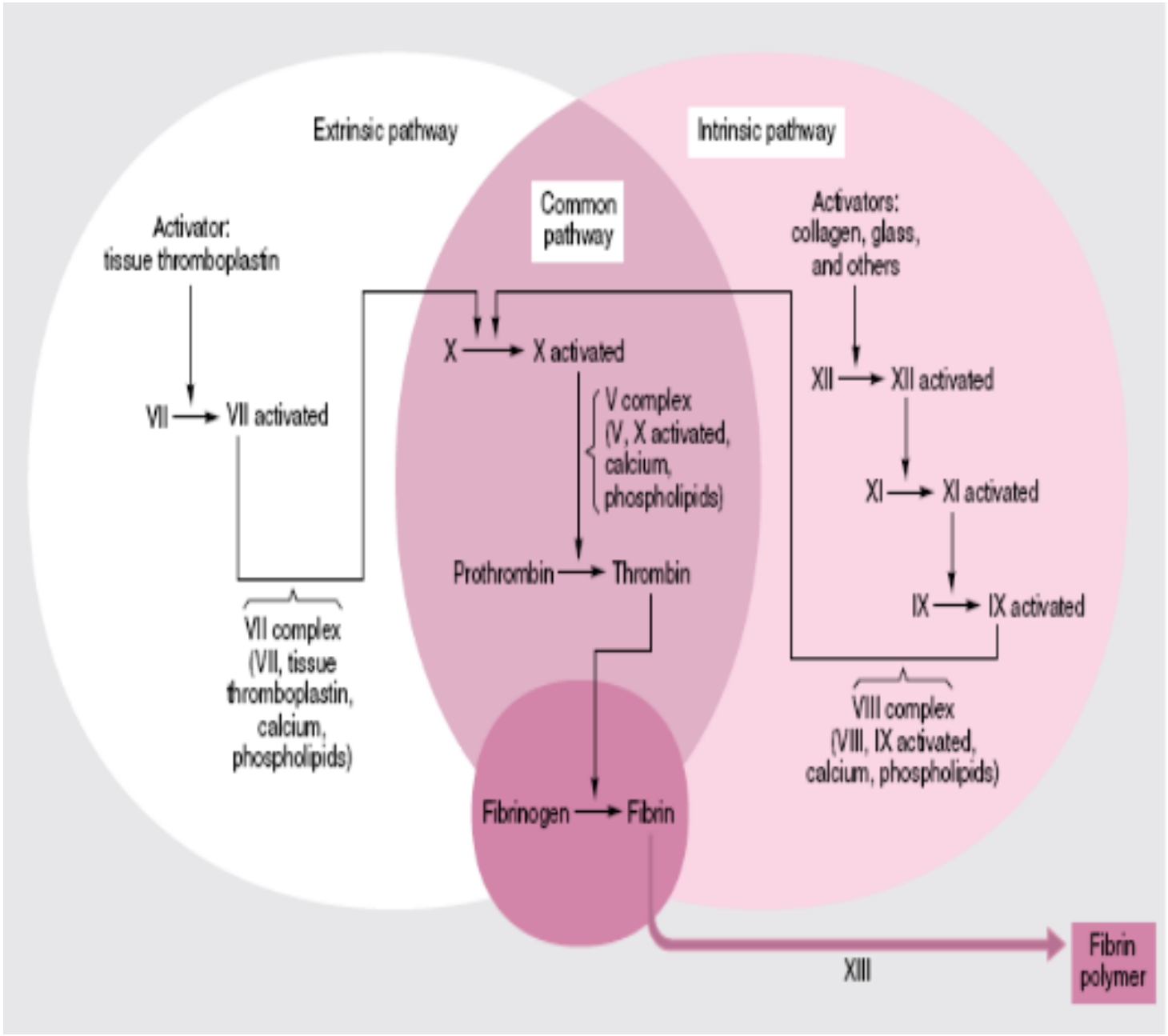
(Definitive plug): Firm

Intrinsic pathway

Extrinsic pathway

Common pathway




Extrinsic pathway
Intrinsic pathway
Common pathway
Activator: tissue thromboplastin
Activators: collagen, glass, and others
VII
VII activated
X
X activated
V complex (V, X activated, calcium, phospholipids)
XII
XII activated
XI
XI activated
IX
IX activated
Prothrombin
Thrombin
VII complex (VII, tissue thromboplastin, calcium, phospholipids)
VIII complex (VIII, IX activated, calcium, phospholipids)
Fibrinogen
Fibrin
XIII
Fibrin polymer



Assessment of haemostatic function

1. Bleeding time.
2. Coagulation tests:
   - Prothrombin time (PT)
   - Activated partial thromboplastin time (aPTT)



Bleeding Time

Definition:

It is the time required to stop bleeding from injured blood vessel by platelet plug formation.

Injury → VC → Platelet plug formation.

Mechanism:

1- VC.

2- Platelet plug formation.

So, it assess both vascular wall integrity, and platelets (number and function).

Done by Filter paper test.



Procedure:

1- Sterile the little finger (the best site is the lobule of ear) with alcohol.

2- Prick with sterile finger-prick.

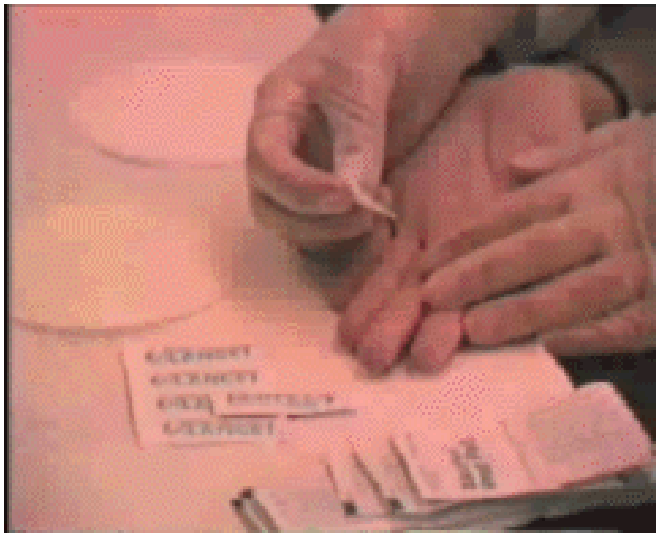
3- Remove the blood drops every **15 sec** by touching the puncture site with filter paper, without pressing or squeezing, repeat till bleeding stop.

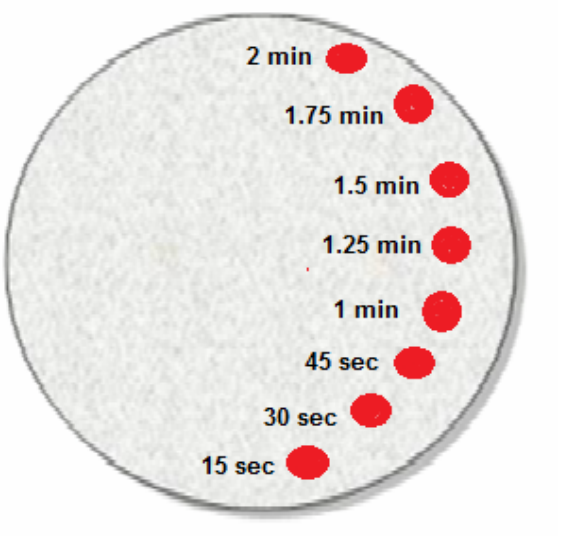
4- Count the number of blood spots on filter paper, calculate the time in min.



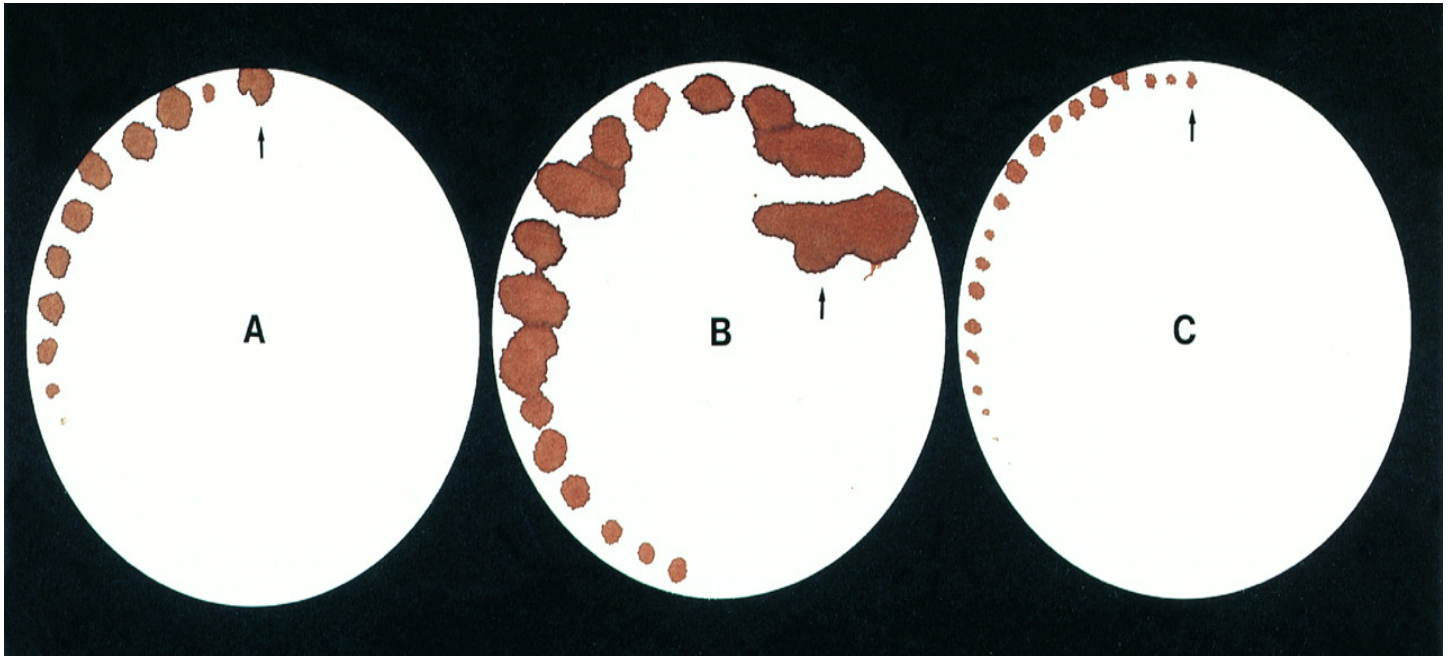







2 min
1.75 min
1.5 min
1.25 min
1 min
45 sec
30 sec
15 sec





Bleeding time = no. of blood drops / 4



Normal Value:

2: 5 min

Prolonged in:

1- Vessel wall defect e.g. Scurvy.

2- Platelet defect e.g. Purpura

Quantitative defect (Thrombocytopenia, i.e decrease no.)

Qualitative defect (Thrombocytopathia, i.e decrease function).







Prothrombin time (PT)

Importance

It is used to test the integrity of extrinsic clotting pathway (factors V, VII, and X, prothrombin, and fibrinogen).

Interpretion of PT test

A prolonged prothrombin time indicates a deficiency in any of factors of extrinsic clotting pathway (VII, X, V, prothrombin, or fibrinogen) as in cases of:

- Vitamin K deficiency
- Liver disease
- Patients receiving a vitamin K-competing drugs coumarin such as warfarin



Activated Partial Thromboplastin Time (aPTT)

Importance

It is used to test the integrity of intrinsic clotting pathway (Factors XII, XI, VIII, IX, X, V, prothrombin, and fibrinogen).

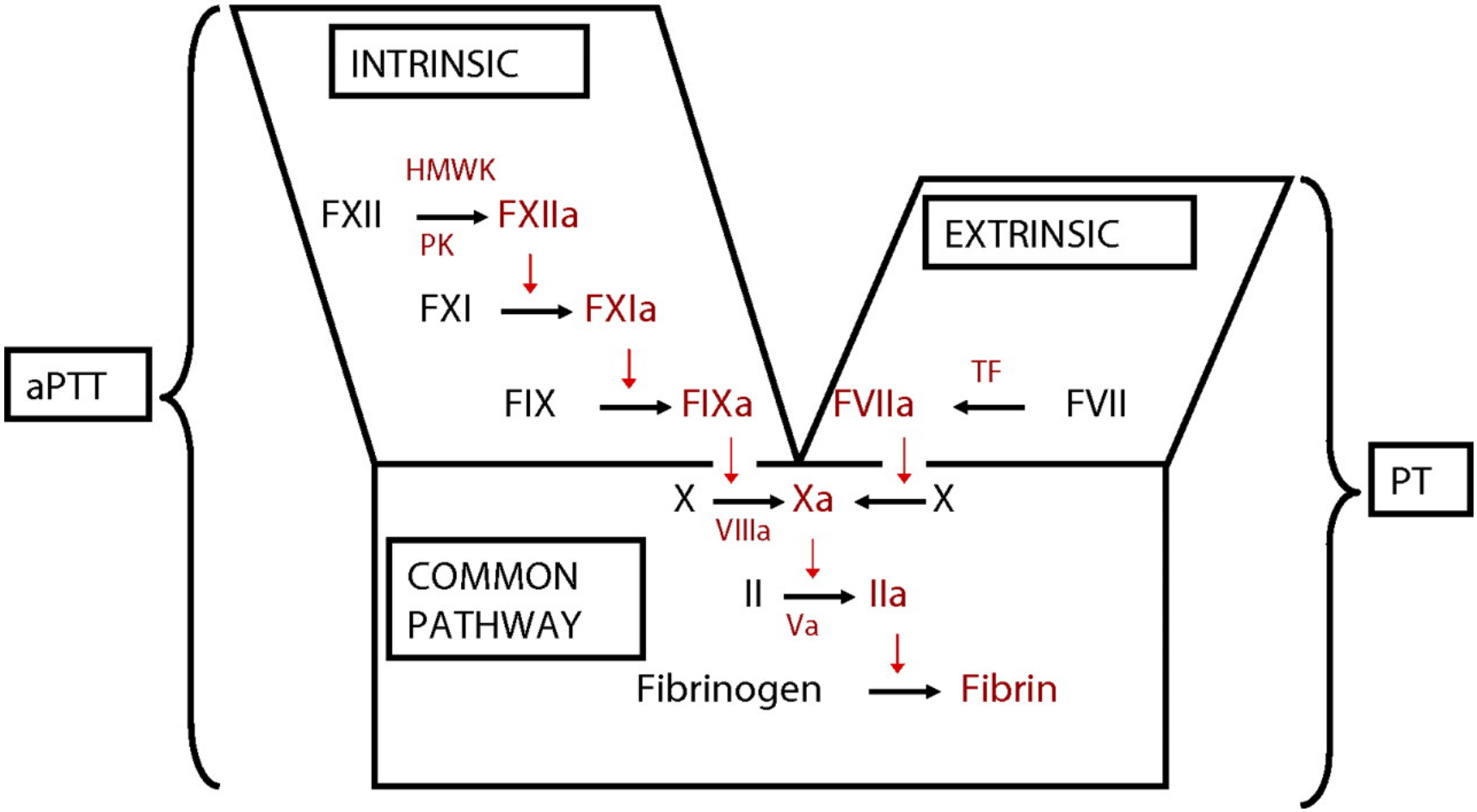


Interpretion of aPTT test

A prolonged activated partial thromboplastin time indicates a congenital or acquired deficiency in any of factors of intrinsic clotting pathway (XII, XI, VIII, IX, X, V, prothrombin, and fibrinogen) as in cases of:

- Haemophilia
- Liver disease
- Heparin therapy

Coagulation Factor Activation




Thank you